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UX Research Portfolio

Updated December 2024

Hello! I'm Tricia

I'm a **mixed-methods researcher** with 10+ years of experience conducting human-centered design and research projects in the creative and educational technology sectors. My superpower is **finding the connections in "messy" data** and **advocating for users** through meaningful recommendations and stories.

I received my **Ph.D in Cognitive Science** from the **University of California, San Diego** where I designed adaptive interfaces for helping novice creators in exploratory thinking.





Research Methodologies

As a mixed-methods researcher, I select methods that are appropriate for the research question, goal, scope, and constraints of the project. Here are some of the methods in my UX toolbox.

- Experimental design
- Literature review
- User personas
- Storyboarding
- Competitive analysis
- Surveys
- In-depth interviews
- Co-design
- Focus groups
- A/B testing
- Iterative prototyping
- Usability testing
- Affinity diagramming
- Statistical & thematic data analysis

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Student perceptions of smart classroom technologies

What do undergraduate students think of smart classrooms (i.e. cameras, microphones, sensors) being used in their classrooms?



Context & Problem Statement.

Smart classroom systems like cameras, microphones, and other sensors make it possible for researchers and instructors to make sense of what's happening in the classroom.

Though their data is collected through these systems, students aren't primary users and don't ever really interact with the systems themselves.

As smart classroom technologies are recent developments, **student perspectives of these systems outside of generalized surveys are not fully understood.**

In this study **I led exploratory, qualitative research** to understand what students thought about smart classroom technology and their impact on learning experiences.



Project Overview.

Team.

- Me: Lead UX researcher
- 2 Junior researchers
 - 4 Faculty project mentors

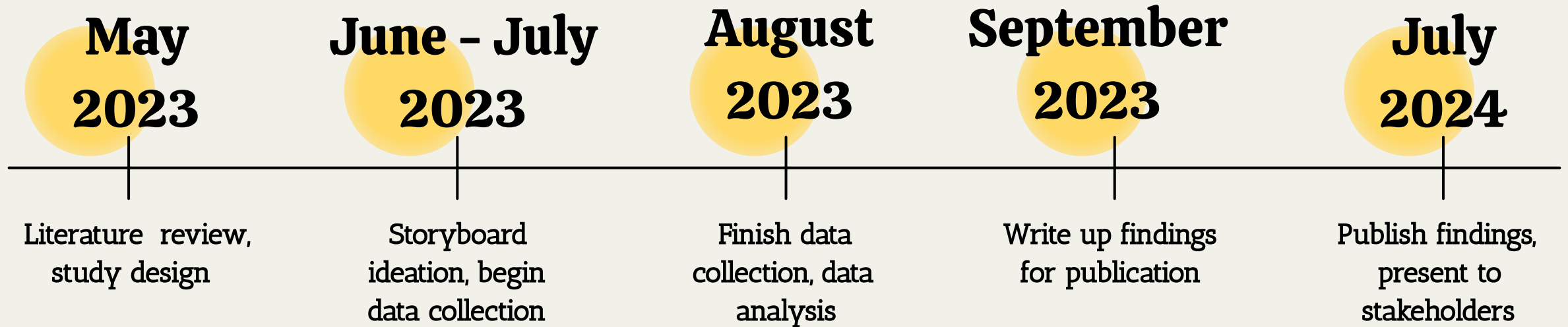
Scope.

- Speed dating with storyboards study with undergraduate students
- Design implications for smart classroom systems

Deliverables.

- 1 Academic publication
 - 2 Presentations to stakeholders

Project Timeline.



Research Methodology.



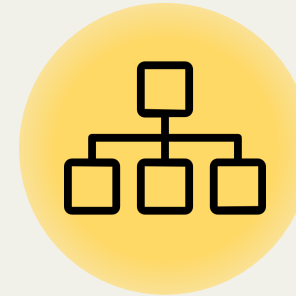
Participants.

14 STEM
undergraduate
students from 11
universities around the
U.S.



Task design.

Speed-dating with
storyboards with 27
storyboards of
potential futures of
smart classrooms.



Data analysis.

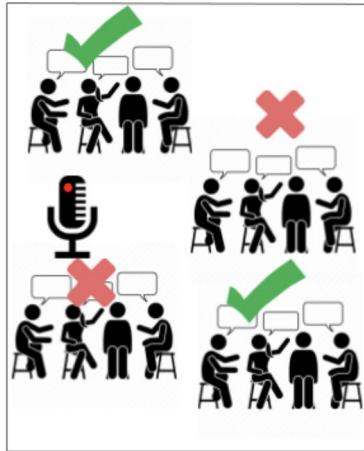
Affinity diagramming
to group emergent
themes together.

Why the speed-dating method?

With any emergent technology, it's difficult to imagine what futures are possible without widespread adoption. Speed dating with storyboards is a method that investigates people's potential acceptance and boundaries with new technologies.



Instructor Alex often has his students break up into small discussion groups in class, but he isn't sure how many students are actually talking about the class material and how many are having unrelated conversations.



Alex uses a sensing system that uses microphones to detect which groups are using relevant keywords in their conversations and which groups are having conversations not about class material.



The sensing system sends this information to instructor Alex in real time so he can see which groups are talking about class material and which are not.

A sample storyboard from our study.



Constraints.

Participants.

Our participants were STEM undergraduate students who may have more technical knowledge and views of privacy than the average student.

Method.

Because speed dating is meant as a future-facing exploratory method, students have had a difficult time sharing thoughts on technology that does not yet exist.

Timing.

We collected data in the summer in order to get a diverse set of undergraduate students from different institutions, which limited the timeline for our study.

Our affinity diagramming unveiled nuanced (and sometimes contradictory) views towards smart classroom technology.

In our speed-dating sessions with students, I uncovered 3 primary themes:

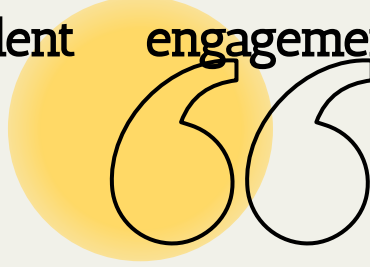
- **Autonomy:** agency and control over one's own learning experience and environment.
- **Connections:** relationships between peers and with the course instructor.
- **Role of technology:** what role technology should (and shouldn't) play in the classroom for both students and instructors.



Students' values & boundaries with smart classroom technology

	Values	Boundaries
Autonomy	Can encourage effort and more participation from students.	Could impede autonomy and lead to students and instructors "gaming the system."
Connections	Want instructors to be proactive in helping students and have an implicit awareness of student needs.	Could lead to disingenuous connections between instructors and students.
Role of Technology	Could provide additional data for instructors to make decisions in class.	Students might feel like "data points" rather than people.

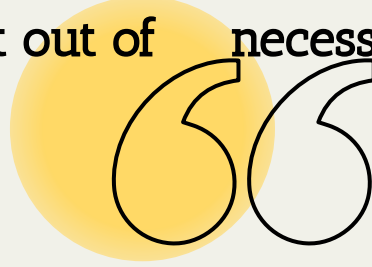
Autonomy: Some students thought smart classrooms could provide an objective view of student engagement. .



It implements more objectivity to [showing effort] because it's not just what the professor kind of internally perceives as people who speak up the most.

— Student participant speaking about smart classroom technologies improving student autonomy and participation.

Autonomy: Other students felt that smart classrooms might lead to disingenuous engagement out of necessity.



If students choose not to take initiative I don't feel the professor should kind of force them to participate...If I don't sit in the front, I know I'm making the conscientious decision that like I'm not ready to be fully engaged, and sometimes I'm pretty okay with that.

— Student participant speaking about potentially feeling forced to engage or participate if smart classroom tech is implemented.

Connections: Smart classrooms can help instructors have implicit awareness of which students need support.



It takes the pressure off of students to like actively say ‘hey I don’t know what’s going on’ and just gently nudge the professor so they can use that information to subtly walk over.

— Student participant speaking about wanting instructors to have an implicit awareness of when students need help instead of needing to ask themselves.

Connections: Some students felt that instructors could already pay attention to all students and wouldn't need further awareness.



In a small class, it's realistically possible for the instructor to see who's participating and who's not so I'm just like questioning the usefulness of such a system...in general, an instructor is able to keep track of 30 students.

- Student participant speaking about smart classroom technologies being unnecessary since instructors can already keep track of up to 30 students.

Role of Technology: Students felt that smart classrooms should provide extra info for instructors to make better decisions in their classrooms.



**It would also be
interesting if [the system] gave them options like ‘here
are five ways that you can improve engagement.’**

— Student participant speaking about smart classrooms providing additional data and strategies for instructors to improve student engagement in class.

Role of Technology: Students opposed how smart classrooms might utilize sensitive or more intrusive data.



It's different if you're looking at someone's attendance because you only need to see...like a hand raise...but with gaze you're analyzing not only where they're looking, but what they're looking at, kind of almost trying to figure out what's going on inside their head.

— Student participant speaking about the data smart classrooms being potentially invasive, particularly when collecting data such as gaze.



Design implications.

Students, instructors, and researchers have **nuanced and sometimes differing views** on learning experiences. These views should be considered in the design of classroom technologies.

Given the sensitive nature of smart classroom data, careful consideration must be made to ensure that the good intentions of technology **do not instead amplify inequities** .

Impact



Academic publication

Published in ACM
Designing Interactive
Systems (DIS) 2023.



Honorable Mention Award

Received award for
being of top 5% of
submitted papers at
ACM DIS 2023.



Presentations to stakeholders

2 presentations to
stakeholders.



CritiqueKit: Adaptive suggestions for giving better feedback

(How) do feedback suggestions help novices give critique on design work?



Context & Problem Statement.

Feedback is important for any learning, especially in domains where there often isn't one right answer, like design. There are far more novices seeking feedback than there are experts who can provide this feedback, which means peer feedback is the most common form of feedback exchange.

However, people have a hard time giving **specific, actionable, and justified** feedback because it isn't an explicitly taught skill.

In this study **I led design, development, and evaluation** of a web application that provides adaptive guidance and suggestions to novices for giving critique on design work.



Project Overview.

Team.

- Me: Lead UX researcher
- 2 Research engineers
- 2 Senior project mentors

Scope.

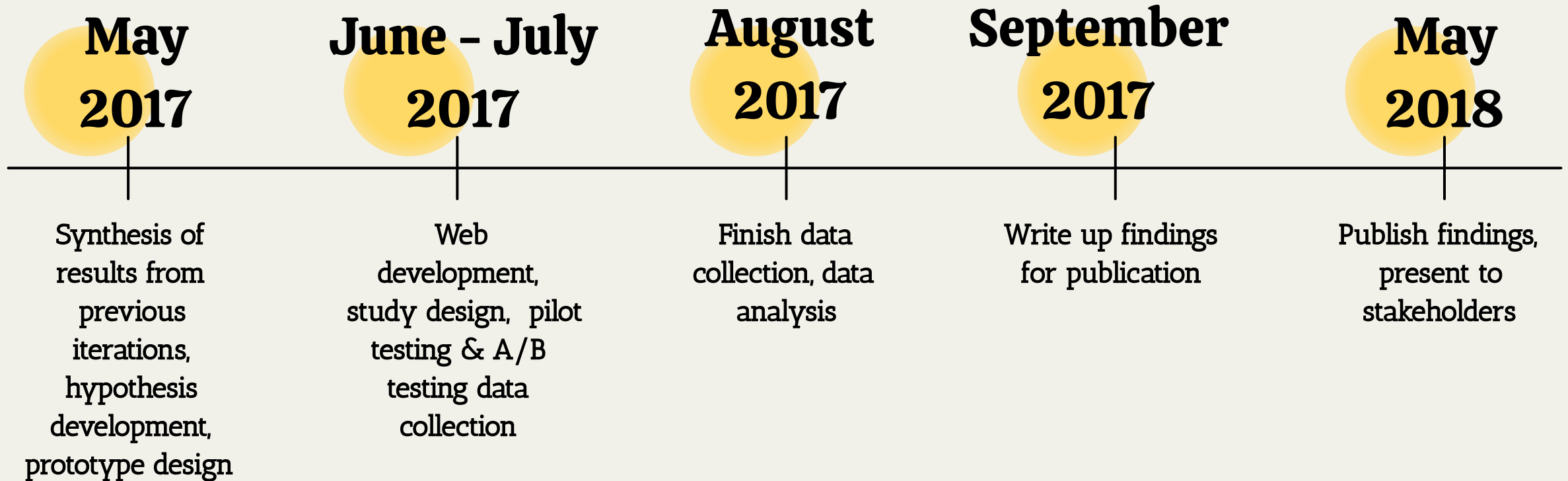
- Web application for providing guidance in giving critique
- A/B testing study to evaluate web app
- Design implications for adaptive systems

Deliverables.

- 2 Academic publications
- 4 Presentations to stakeholders

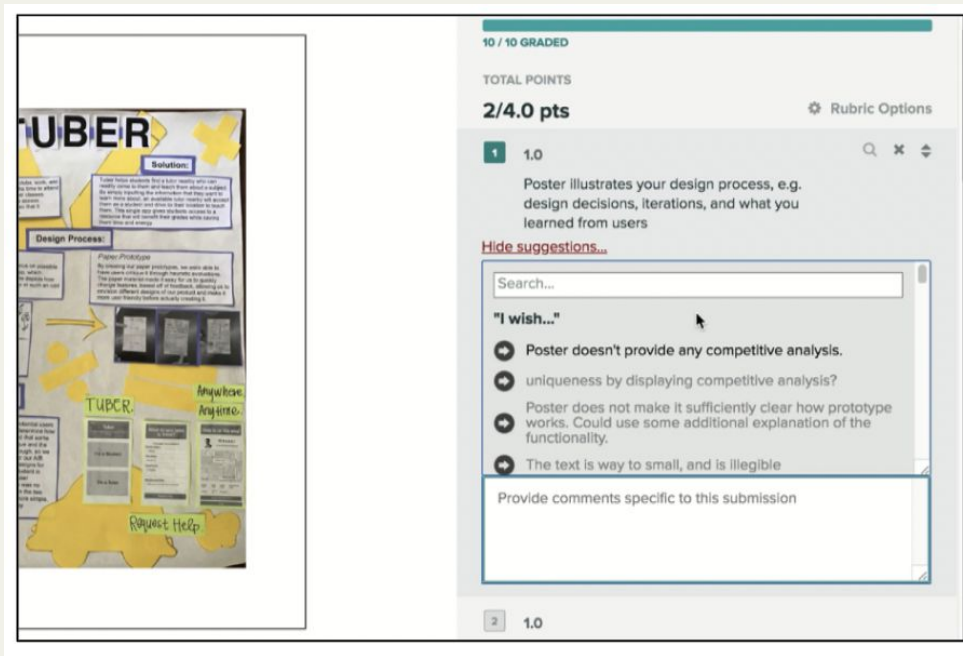
Project Timeline.

The entire span of this project spans 1.5 years. This case study will cover a 1 year span where I led design and research from study conception to completion for our final design of CritiqueKit.



Iterative Prototyping of CritiqueKit

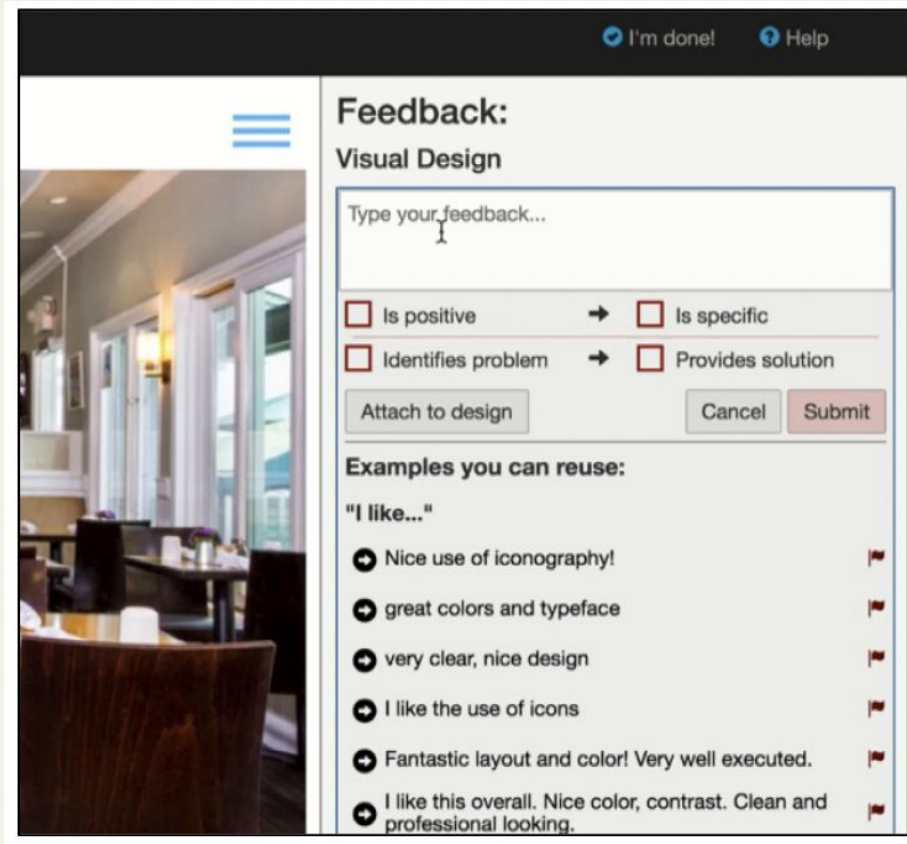
Over a year, CritiqueKit's interface and features evolved based on iterative user feedback and evaluations. Here, I provide a brief overview of the previous CritiqueKit designs and lessons learned from them.



We first developed CritiqueKit as a browser extension for the Gradescope grading tool where users could see crowdsourced examples of feedback comments and select them to use in their own feedback comments.

From an **online deployment** of CritiqueKit with 8 teaching assistants in a 35-person undergraduate design studio course, we gathered exploratory data of how people use feedback examples.

Iterative Prototyping of CritiqueKit



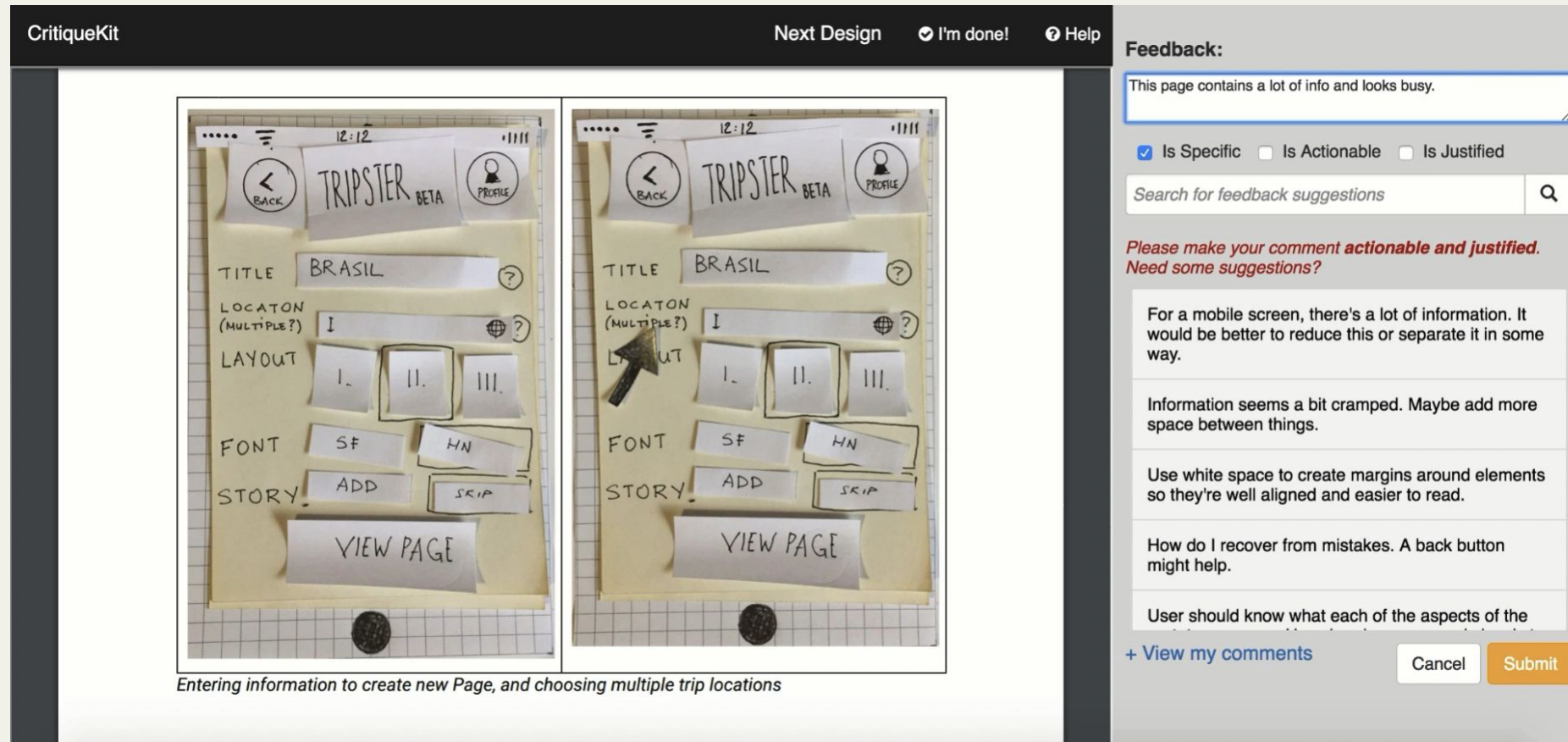
From our deployment with CritiqueKit v1, we made CritiqueKit a standalone web app and added 2 features to better support novice users:

- a panel of checkboxes that update based on the characteristics of the feedback comment
- a box of suggested examples that users can select to use in their own feedback comments.

From an **online deployment** and **online A/B testing experiment** with 69 undergraduate design students, we discovered that users found the checkboxes and the examples unhelpful, likely because students have less experience with design concepts and how to apply the examples.

CritiqueKit: Final design

Key Insight: In our previous versions of CritiqueKit, the examples were not curated and remained the same for all users. What if the examples could adapt to the user?



CritiqueKit: Final design changes

From feedback from the initial versions of CritiqueKit, I made two major changes to the final design:

- 1) Categories in the guidance panel changed to "**Is Specific**", "**Is Actionable**", and "**Is Justified**" to encourage users to provide reasoning to their critique
- 2) The examples within the suggestions box **adapt** based on how a critique comment is categorized in the guidance panel

Interactive guidance panel to categorize critique

Adaptive suggestions based on guidance panel

The screenshot displays the CritiqueKit interface. At the top, a 'Feedback:' section contains a text input field with the placeholder text 'This page contains a lot of info and looks busy.' Below this input field are three checkboxes: 'Is Specific' (checked), 'Is Actionable' (unchecked), and 'Is Justified' (unchecked). A search bar labeled 'Search for feedback suggestions' is positioned below the checkboxes. The main content area, titled 'Please make your comment **actionable** and **justified**. Need some suggestions?', lists four adaptive suggestions based on the selected 'Is Specific' category. These suggestions are: 'For a mobile screen, there's a lot of information. It would be better to reduce this or separate it in some way.', 'Information seems a bit cramped. Maybe add more space between things.', 'Use white space to create margins around elements so they're well aligned and easier to read.', and 'How do I recover from mistakes. A back button might help.' At the bottom of the form, there are three buttons: '+ View my comments' (blue), 'Cancel' (white), and 'Submit' (orange).

Research Methodology.



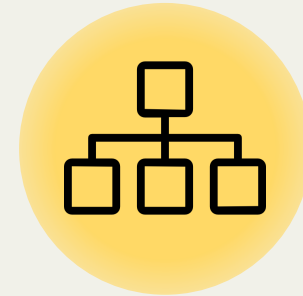
Participants.

47 undergraduate students who were novices in design.



Task design.

Provide critique on 2 paper prototypes in the CritiqueKit interface and engage in a post-study interview.



Data analysis.

Quantitative analyses to compare the number of critique comments that were categorized as Specific, Actionable, and Justified.

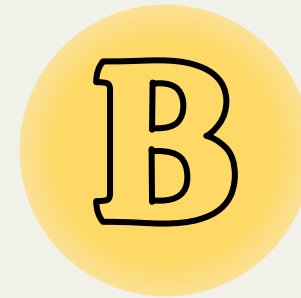
Study design: A/B Testing

Hypothesis: Adaptive suggestions and guidance will lead to more specific, actionable, and justified critique.



Experimental condition (n=24).

Participants provided feedback in the CritiqueKit interface with the guidance panel and adaptive examples.



Control condition (n=23).

Participants provided feedback in an identical CritiqueKit interface without the guidance panel or adaptive examples.



Constraints.

Budget.

We didn't have a budget to pay participants so we were limited to undergraduate students who could receive course credit from participating in research.

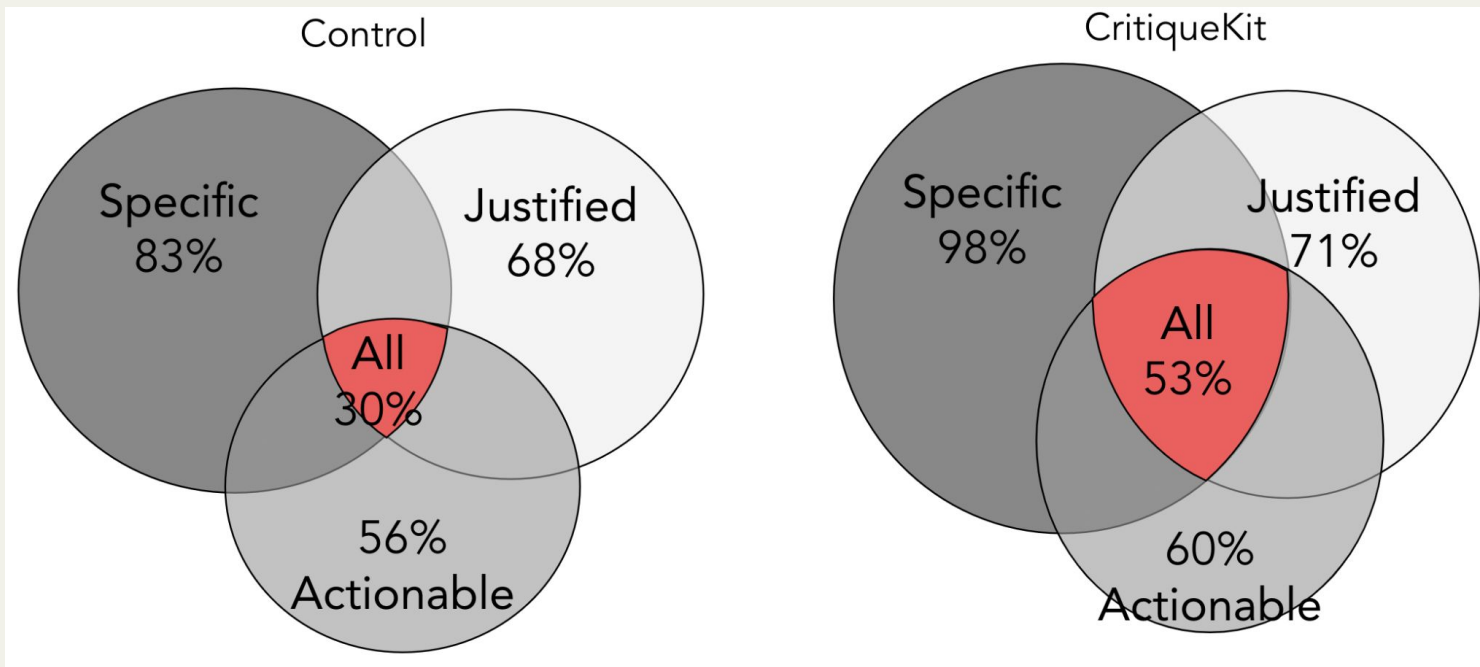
Sample.

Our limitation to an undergraduate student population meant our students were mostly social science students in courses that required research participation credits..

Timing.

Running our studies with students meant following the 10-week academic quarter system, which put constraints on our design and development timelines.

Key Findings



Participants who used the experimental version of CritiqueKit gave more feedback that was **specific, actionable, and justified** than participants in the control condition.

In the post-study interview, participants **found the suggestions to be helpful** : 4.29 / 5 mean rating. However, they were mixed in their ratings of helpfulness for the guidance panel: 3.67 / 5 mean rating.

Key Findings: Experimental participants felt the suggestions helped put words to their thoughts, though we also noticed some participants over-relying on the suggestions rather than using them as a starting point for their own thoughts.



I was a bit lost at first because I didn't know how to describe my thoughts. The suggestions helped me figure out how I should describe what I was thinking.

Key Findings: The guidance panel helped participants better structure their feedback.



**Because I don't know how to give feedback, [the categories]
give me a criteria.**

Key Findings: Participants also sometimes felt the guidance panel was inaccurate at times.



It felt quick to judge, it felt like it wasn't reading what I was saying.



Design implications.

Examples are not enough. Adaptive interfaces that present examples or suggestions to users should provide **relevant** examples that fit into the user's **context**.

For examples to be useful outside a particular context or individual work, they should be based on **structural** features or concepts that can be applied in multiple contexts.

Impact



Usage

CritiqueKit has been used by over 400 students and over 100 instructors and teaching assistants.



Academic publications

Published in ACM Human Factors in Computing (CHI) 2018 and ACM User Interface & Software Technology (UIST) 2017.



Honorable Mention Award

Received award for being of top 5% of submitted papers at ACM CHI 2018.



Presentations to stakeholders

4 presentations to stakeholders.



EPiC: An interactive online tool for assessing children's patterning abilities

Adapting an in-person developmental psychology task to an online version in response to the COVID-19 pandemic.



Context & Problem Statement.

The COVID-19 pandemic meant in-person lab studies needed to be paused or moved to an online task. For developmental psychology research, this posed a real challenge because young children (ages 3-6 years old) are often developmentally unable to do long, unmoderated tasks on a digital device.

Most online tasks for young children often consisted of brief, simple tasks, but **longer and more complex tasks had not yet been explored in online contexts**.

In this study **I led prototyping and a deployment** of an online version of a complex, in-person patterning task for young children.



Project Overview.

Team.

- Me: Lead UX researcher
- 3 Junior researchers
- 1 Senior project mentor

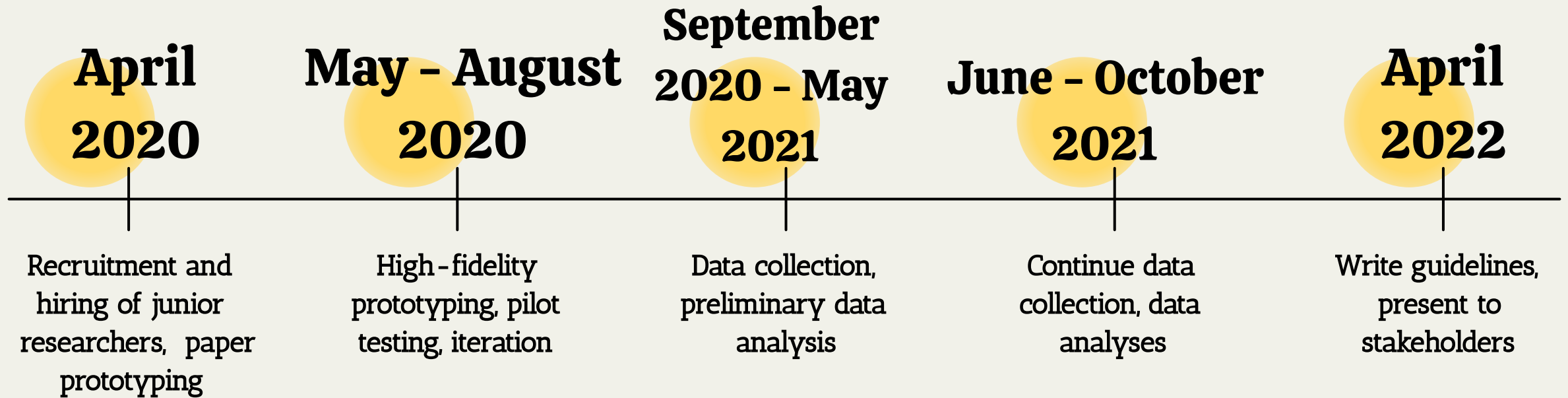
Scope.

- An interactive prototype to adapt an in-person patterning task online
 - Results from a deployment of the prototype
- Design implications

Deliverables.

- Guidelines for adapting developmental psychology tasks online
- 2 Presentations to stakeholders

Project Timeline.



Design needs

To design an interactive prototype suitable for children, their parents, and the researchers conducting the study, I considered the following design needs for these stakeholder groups.



Children

Children need to be able to manipulate stimuli without direct assistance and interact with stimuli on any device (*i.e.* phone, laptop, tablet).



Parents

Parents need to be able to access the technology easily and allow children to independently complete the tasks.

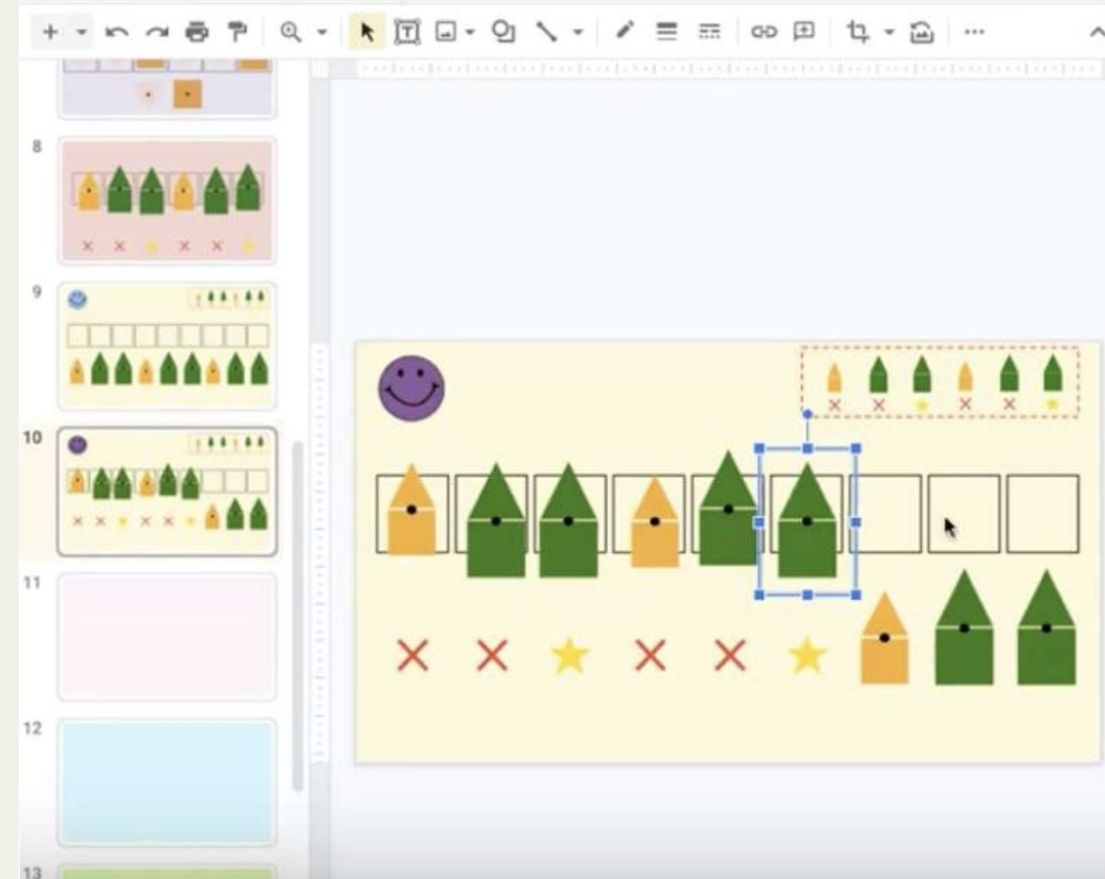


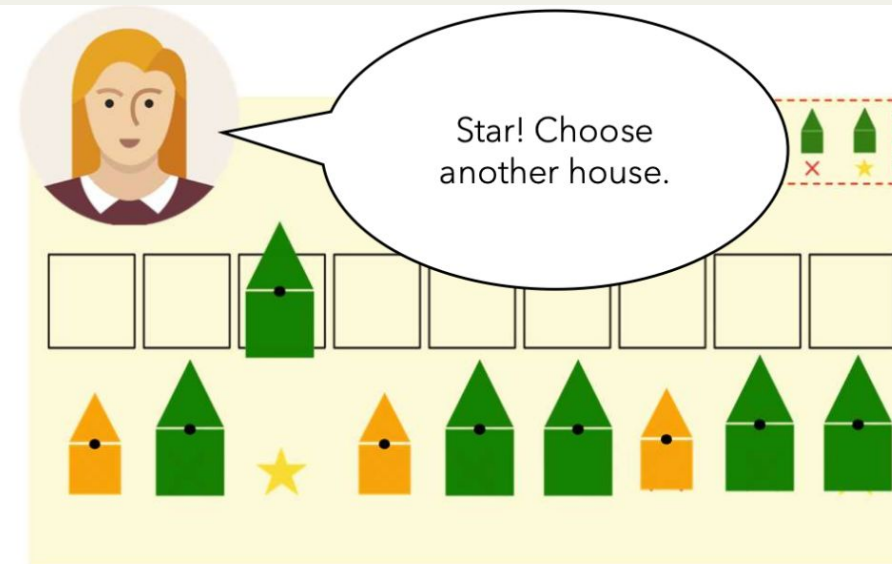
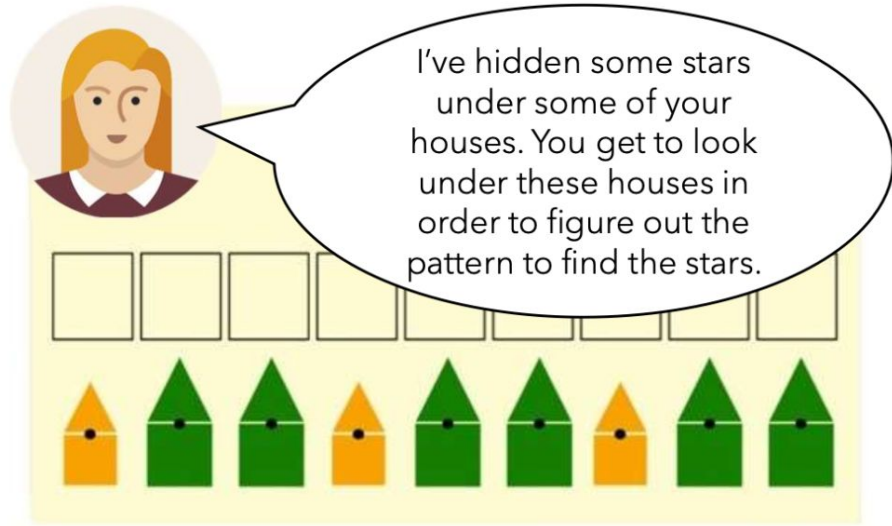
Researchers

Researchers need to be able to see how children perform in real-time and make sure children work consistently in a controlled environment.

Google Slides as a platform for interactive prototyping

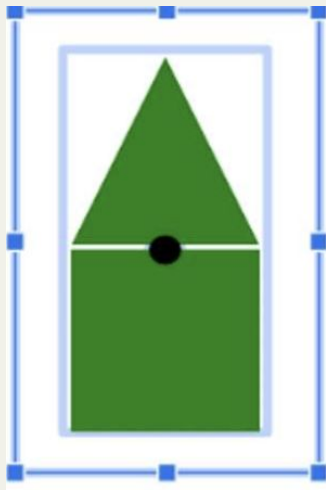
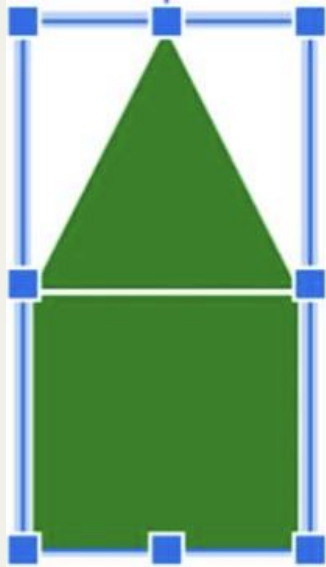
After a competitive analysis with other online platforms, I chose Google Slides because it doesn't require additional software downloads or an account and the researcher and participant can see real-time, synchronous changes.





EPiC: Exploratory Patterning in Children

In our Google Slides prototype, we adapted a "Find the Stars" patterning game where children would explore the pattern to learn where stars were hidden underneath shapes.



Key design decisions: Easily manipulable stimuli

In our pilot testing, we found that children had difficulty with the **precision needed to select shapes** on a touchscreen or with a mouse.

To mitigate this challenge, I added an **invisible bounding box** around the shape to make the surface area for clicks larger. I also added a black button to each shape to direct children to touch a specific part of the shape.

Key design decisions: Transitions

Our pilot data also showed that some young children did not consistently know numbers or the difference between left and right.

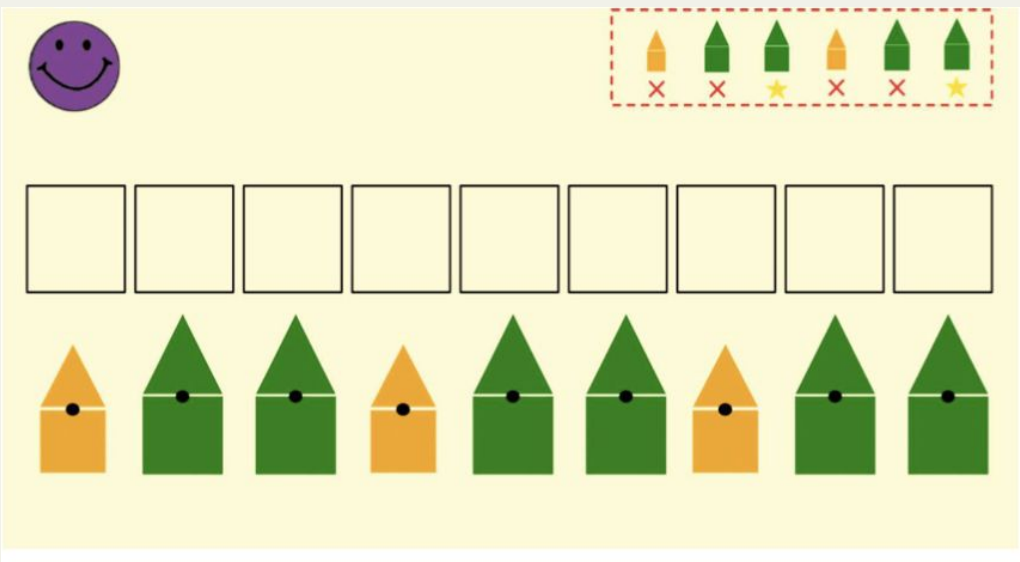
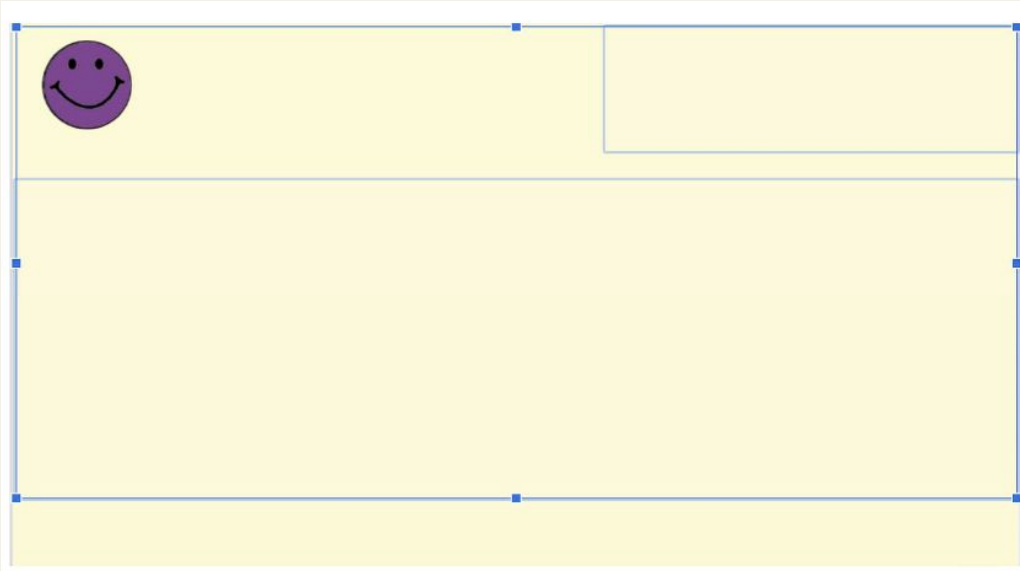
To help children transition between slides in our prototype, the research script included the use of **colors** rather than numbers or directions to help children navigate through the task.



Key design decisions: Scaffolding attention

From our pilot data, we found that children wanted to interact with the shapes before being instructed to do so and did not know where to direct their attention as pointing gestures are not easily seen online.

I added shapes to **hide and reveal stimuli** at the right times to direct children's attention to the appropriate areas.



Deployment Details.



Participants.

280 children aged 5-6 years old with 52 exclusions, leaving a remaining sample of 228 participants.



Task design.

The "Find the Stars" game where children explored shape patterns to find hidden stars.



Data collection.

All studies conducted through Google Slides and Zoom for screen recordings.



Key Findings

Though some participants were excluded due to parental involvement, some parental guidance, such as **neutral prompting or encouragement** , were helpful, especially if the child had difficulty with the technology.

Some children still had trouble with **drag-and-drop mechanisms** , which led to the researcher moving stimuli based on the child's selection.

Children enjoyed the **playful and immersive** nature of the patterning game.



Design implications.

Our prototype was designed for synchronous research where the researcher could assist children if they had difficulty with the task or technology. For asynchronous interactive technologies, the interface should provide **simple, developmentally - appropriate input mechanisms** for the child to engage.

Children are **capable of completing complex, interactive tasks** provided they have appropriate guidance, immersion, and encouragement.

Impact



Usage & Accessibility

EPiC increased accessibility to research with 280 participants in 13 months compared to 90 children for the in-person task in the same time frame.



Guidelines

Wrote guidelines for online research and creating online tasks with children



Presentations to stakeholders

2 presentations to stakeholders.

Thank you!

Do you have any questions?



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